Flying-wing QuadPlane | Attitude review

Compare commanded and measured attitude across pitch, roll and yaw.

PID Review Tool

Pitch : ATT.DesPitch (blue) vs ATT.Pitch (orange)

The orange (actual pitch) does follow the blue (desired pitch), but it is **noisy and wobbly**



Stick Inputs are Small
Desired pitch (blue) only moves about ±2°.
But the actual pitch (orange) swings **much further** (up to ±6°).
This confirms **the controller is overreacting** to even small inputs.

Overshoot & Oscillation
- Similar to roll, the pitch overshoots the command and rings before settling.
- That means **Pitch P-gain is too high**.

Tracking
- The orange (actual pitch) does follow the blue (desired pitch), but it is **noisy and wobbly**.
- Especially after ~14:20:00, the orange line overshoots and oscillates.

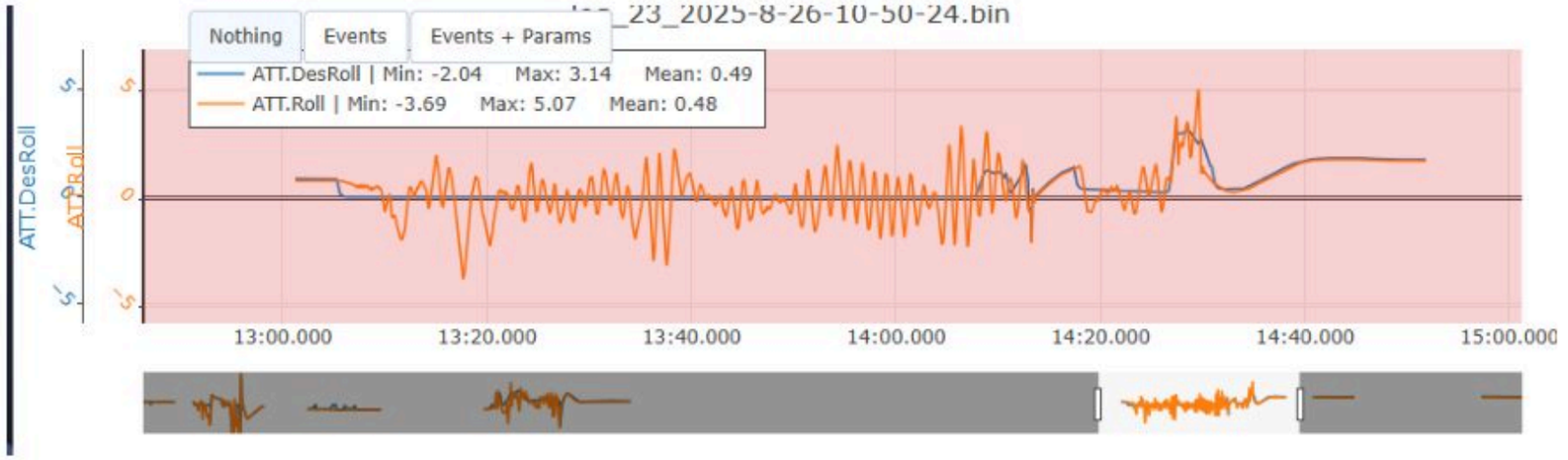
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PID Review Tool

Roll : ATT.DesPitch (blue) vs ATT.Pitch (orange)

Aggressive oscillations (orange line)
Your **ANG.Roll** is very “jagged”, meaning the controller is fighting too hard → **P-term too high**.



- Reduce **Q_A_RAT_RLL_P** by ~40%.
- 0.20 → set to 0.12.

Overshoot
When you command a roll (blue line steps up), the actual roll (red) **overshoots strongly** and then oscillates before settling.
This is a textbook case of **too much RATE_P** or too much ANG_P gain.

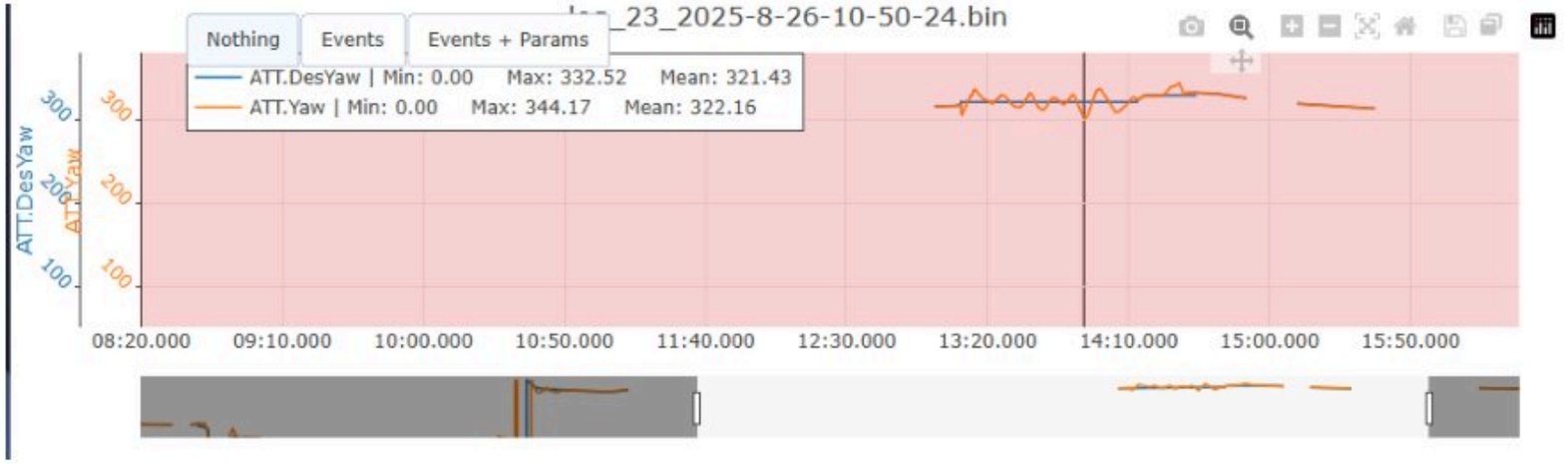
Des vs Actual
Desired is only about ±3° roll, but the actual oscillates much more violently.
That means the **flight controller is amplifying your small stick inputs**.

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Compare commanded and measured attitude across pitch, roll and yaw.

PID Review Tool

Yaw : ATT.DesYaw vs ATT.Yaw



Alignment

When DesYaw moves, Actual Yaw tracks it **without major delay or overshoot**.
This means **Yaw PID is okay**.

Overshoot & Oscillation

- Similar to roll, the pitch overshoots the command and rings before settling.
- That means **Do NOT touch yaw PIDs for now** (Q_A_RAT_YAW_P, I, D)..

Actual Yaw (orange)

It's fairly stable, only small oscillations (~1–2°).

No big overshoot or hunting like Roll/Pitch.